Recursive Function

What is Recursive function?

- When a function calls itself, then the function is called recursive function.
- A recursive function creates a process of chaining.

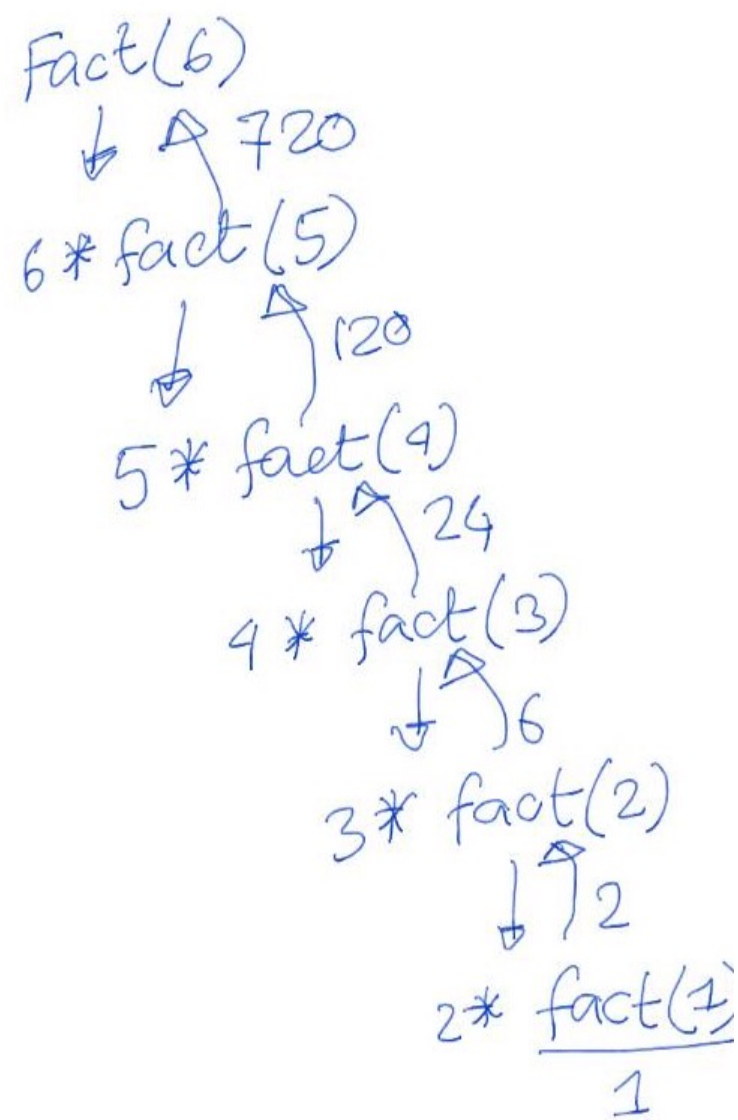
Steps of Recursive Function

- Termination
- Reduction

Example 1 (Factorial)

```
#include <stdio.h>

long Factorial(int n);     /* function prototype */

main(){
     int  n;
     long  fvalue;

    printf(“Enter a value: “);
    scantf(“%d”, &n);

    fvalue = Factorial(n);     /* calling function */

    printf(“The factorial of %d is %ld.”, n, fvalue);
    return 0;
}

long Factorial(int n){        /* function definition */
    if (n==1) return 1;
    return n*Factorial(n-1);
}
```

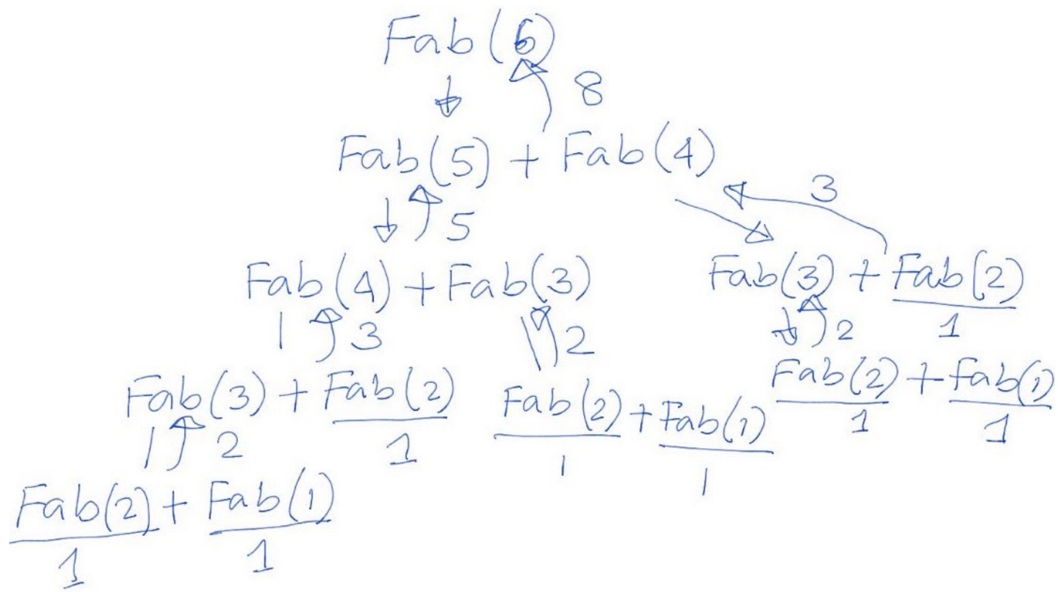
Chaining Process of Recursive Function



Example 2 (Fibonacci Number)

1 1 2 3 5 8 13 21 ……

Output of Recursive Function



Recursive Definition:

1. Termination,

$F_1 = 1$ **and** $F_2 = 1$

2. Reduction.

$F_n = F_{n-1} + F_{n-2}$

```
int Fab(int n){
    if (n == 1 || n == 2) return 1;
    return (Fab(n-1)+Fab(n-2));
}
```

Example 3 (GCD)

Find gcd of 15 and 9.

```
9 )  15  ( 1
      9
   ---------
      6) 9 ( 1
         6
      ---------
         3 ) 6 ( 2
             6
           ------
```

```
int GCD(int a, int b){
    if (a % b == 0) return b;
    return GCD(b, a%b);
}
```

Find gcd of a and b

Recursive Definition:

1. **Termination,**

   if (a % b == 0) return b;

2. **Reduction.**

   return gcd(b, a%b)

Practice

Problem 7.11(a) (Page 7.37)

The Legendre polynomials can be calculated by means of the formula

$P_0 = 1, P_1 = x$

$P_n = [(2n-1)/n]xP_{n-1} – [(n-1)/n]P_{n-2}$

Where n = 2, 3, 4, ….. and x is any floating-point number between -1 and 1. (Note that the Legendre polynomials are floating-point quantities). Write a recursive function to generate nth Legendre polynomial term. Let the values of n and x be input parameters.

```
main(){
    ………………………………
}
```

```
float LP(int n, float x){
    ………………………………
}
```